MIL-P-63515A (AR)
AMENDMENT 2
10 October 1986
SUPERSEDING
AMENDMENT 1
18 July 1986

MILITARY SPECIFICATION

PROPELLANT M30 FOR USE IN CARTRIDGE
105MM, APFSDS-T, M833

This Amendment forms a part of Military Specification MIL-P-63515A (AR) dated 1 September 1983 and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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* 3.1.2, Table I: Delete "18.0 ± 2.0" and substitute "20.0 ± 2.0".

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4.5.2, Line 1: Add "(See 4.5.6)" after "Proving ground assessment".

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Add the following new paragraphs:

"4.5.6 Loading, assembling, and packing (LAP) for ballistic tests. Ten propellant containers per lot marked for ballistic samples shall be placed in an explosive holding area for twenty-four hours prior to unsealing so that the temperature of the contents will be at the same temperature as the loading area. Each round shall be assembled in accordance with drawing 9342932 and the following:

a. Each round shall be assembled with calibration components and two M11 gages with a gage clip to retain the gage in place at the base of the case.

b. Each round shall be marked with the propellant lot number, propellant container number, charge weight, projectile weight, M11 gage number, and test temperature.

LAP of rounds for propellant tests shall be done in two phases, charge establishment and uniformity.

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4.5.6.1 Charge establishment phase. This phase shall consist of nine rounds with one round loaded from each propellant container with one container unused. Three rounds shall be loaded at 95% of the service charge weight. Three rounds shall be loaded at 100% of the service charge weight and three rounds shall be loaded at 105% of the service charge weight. After sufficient information has been obtained about the propellant charge weight, velocity, and pressure relationships, loading may change to the constant slope method for the charge establishment phase. During use of the constant slope method of loading and testing, three rounds instead of nine will be assembled at the same charge weight from three different propellant containers at the expected service charge weight. Charge weights for the charge establishment phase will be provided by AMSMC-QAM-A, Rock Island, IL, or AMSMC-QAR-R, Dover, NJ.

4.5.6.2 Uniformity phase. This phase shall be loaded from the same ten propellant containers used for the charge establishment phase with one round assembled from each propellant container for each temperature phase. A calculated charge weight shall be used for the uniformity rounds and will be provided by the proving ground based on the test results of the charge establishment phase. The quantity of complete rounds for the uniformity phase shall be assembled in accordance with the following.

4.5.6.2.1 Preproduction and initial production. Preproduction shall consist of any pilot lots and initial production shall be the first five production lots and every fifth production lot thereafter. Thirty rounds shall be loaded with the calculated charge weight with three rounds assembled from each propellant container. Each of the three rounds from one propellant container shall be marked -35°F, 70°F, or 125°F. Rounds marked for -35°F and 125°F testing (twenty total) shall be drilled for adaption of differential pressure instrumentation. A template shall be used for drilling these holes such that the holes in the cartridge case will align with the holes in the gun tube.

4.5.6.2.2 Subsequent production.

a. After the initial production series is completed every fifth lot shall be loaded in accordance with the schedule of paragraph 4.5.6.2.1, initial production.

b. The remaining lots will be assembled for subsequent production testing (20 round sequence). Twenty rounds shall be assembled with the calculated charge weight with two rounds assembled from each propellant container. Each of the two rounds assembled from the propellant container shall be marked 70°F or 125°F."

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